

Trenchline (Coastal / Defensive / Fortifications & Defensive Operations)

Executive Summary

This document outlines the design and implementation of a defensive trenchline system, encompassing coastal defense, layered fortifications, and automated coastal operations. The system features offshore bunkers, beach obstacles, trench networks, artillery batteries, hardened pillboxes, coastal roads, and inland fallback lines. The main embankment (Obstacle Protection) is autonomous defense bats, sensor-network routing, control-relay warfare, and statistics under sustained use conditions. In AI Neural (AntiNet), the main anchors at "4-5 m" average distance, covering a 10x10 m physical area representing "1 km²" of coastal defense terrain. Key features include trench networks, beach chokepoints, offshore overwatch, and multi-layer defense lines.

Design Goals & Gameplay

Layered Coastal Defense

Trenchline positions multi-ring defense: beach obstacles, trench networks, pillboxes, and inland bunkers. Attackers must breach sequential layers while defenders coordinate autonomous countermeasures.

Snap & Breach Mechanics

Artillery, naval bombardment, drones, and armored vehicles play major roles. Trenches, bunkers, and pillboxes can be destroyed or partially collapsed.

Autonomy & Defensive Coordination

Autonomous units prioritize (strategize), follow-back positions, counter-battery fire, and overwatch from offshore. Units adapt to breached trenches, destroyed bunkers, and shifting defensive priorities.

Sensor & Information Warfare

Sea spray, smoke, flares, and off-occlusion distort sensors. Relay towers on offshore restore clarity. Destroying relays creates blind pockets across the defense line.

Logistics & Resources

Fast deployment, air-lift, and coastal roads serve as supply nodes. Convoys must navigate interior routes and avoid ambush-prone beach corridors.

Deployment & Progress

A hard deployment budget (~10,000 points) forces players to choose between siege units, defensive armor, or autonomous infantry. Blue deploys from the southeast coastal trenchline, Red deploys from the northwest inland command bunker.

Spatial Constraints (Nreal AR)

Device Capabilities

Screen: 1080p, 120° diagonal FOV, 1800x1080 per-eye resolution. Optional viewing distance: ~4-5 m.

Playable Area & Scaling

Location: physical area: 1 km². Trenchline coastal zone (0.5 km scale). Designed for desktop AR.

Tracking & Occlusion

Real-time tracking across horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view, assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid, 0 to 1 m. Origin (0,0) = southeast corner.

Key Locations

Blue HQ Coastal Trench Command: ~ (0,0), 150x150 m Trenches, sandbags, pillboxes, armored coastline.

Red HQ Inland Command Bunker: ~ (1000,1000), 150x150 m Hardened bunker with radar mast, AA emplacements, and artillery.

Coastal Trench Network: ~ (500,500), 100x100 m Trenches, defensive zone trenches, fire arcs, bunkers.

Coastal Battery Base: ~ (500,800), 120x120 m Artillery units, fuel tanks, repair bay.

Offshore Overwatch Line: ~ (200,300), 100x100 m High-ground overwatch, ideal for sensors and long-range fire.

Beach Landing Corridor: ~ (0,400) - (500,400) Primary amphibious chokepoint, layered defense and obstacles.

Inland Road Base: ~ (1000,0) - (500,0) 400x100 m Primary overwatch point with aerial cover.

CIF Relay A: ~ (800,100), 120x120 m Long-range sensor coverage point.

CIF Relay B: ~ (100,900), 120x120 m Coastal radar and counter-battery node.

Naval Base Bunker: ~ (0,1000) 100x100 m Primary coastal zone.

Chokepoints & Sightlines

Beach Landing Corridor: Primary chokepoint, attackers must breach obstacles and pillboxes.

Central Trench Network

Interior foraging routes, extremely strong defensive positions.

Offshore Overwatch

Long-range sightlines, ideal for artillery and sensors.

Inland Road Route

Practical but safe, vulnerable to overwatch from offshore.

Sightlines

CIF Relay A provides long-range sensor coverage (~200 m). Sea spray and smoke create fog-of-war pockets.

Deployment Zones

Blue Deployment: 100 coastal autonomous defense stations.

Red Deployment: 50 inland command bunker + artillery.

Naval Deployment: 200 offshore for defense groups.

Assets & Environment

Terrain

Beaches, offshore, trenches, bunkers, pillboxes, coastal roads.

Vegetation

Sparse shrubs, grass patches, coastal vegetation.

Buildings & Props

Military: AA guns, radar dishes, bunkers, pillboxes. Industrial: Generators, pipelines, fuel tanks. Logistics: Trucks, crates, fuel drums, drones.

Effects

Smoke, sparks, dust, explosions, sea spray.

Audio

Artillery, waves, machinery, alarms, radio chatter.

LOD & Budget

Asset-Cost Breakdown Table

Asset/Category	Count	Volume	Total Cost
Trenches	3	200k	600k
Offshore	4	10k	120k
Batteries	15	10k	750k
Beach/Fire	4	10k	120k
Props	50	10k	500k
Total	—	—	~2.3M

Texture Budget

2048K for terrain/buildings, 1024K for props, 1024K for compression. Target: 4000 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal 4.27 (GDK) + Nreal SDK (ARKit2) + ARCore/ARKit2 Core modules.

Render Pipeline

Forward rendering, stationary lights, baked GI, aggressive culling.

Anchor

Single-point anchor for entire map.

Interaction

Phone gesture for pre-baked, VR2 as screen-space widgets.

Multilayer

Overlays and HUDs, off-screen HUDs to camera.

Implementation Plan & Testing

Milestones

1. Prototype Layout: ~ Month 1. Build-out trenchline, offshore, beach corridor, AA anchor test.

2. Core Systems: ~ Month 2. AI/AR, defense, autonomous behaviors, base logic.

3. AR Integration: ~ Month 3. ARKit2, ARCore, phone integration, base interactivity.

4. Content Fill: ~ Month 4-5. Final models, textures, lighting, audio.

5. Deployment & Test: ~ Month 6-7. Deployment, testing, ARKit2/ARCore, ARCore/ARKit2.

6. Testing & QA: ~ Month 8. Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify defensive logic, fallback behavior, breach zone navigation.

Comms/Sensors

Test relay distribution, fog-of-war updates, LOS checks.

Performance

Address render on target device.

QA Checklist

Navigation, logic logic, deployment budget enforcement, anchor drift, sensor accuracy.